STAINLESS STEEL MANUFACTURING · STEEL MELTING SHOP

AOD Operations

Process Analysis, Standardisation and Business Requirements

Portfolio edition. This is a sanitised version of work carried out in an operations role at a large stainless steel producer. The employer is not named, and all process parameters, quantities, temperatures, gas ranges, equipment identifiers and timings shown are illustrative placeholders, not operating values. The analysis structure, business rules, requirements, controls and validation logic reflect the work as performed. Nothing commercially sensitive is reproduced.

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About This Document

This document consolidates three pieces of process and requirements work carried out in the AOD (Argon Oxygen Decarburisation) area of a Steel Melting Shop. Each part addresses a distinct operational problem, and each follows the same structure: business problem, objectives and scope, stakeholders, current-state analysis, future-state design, rules and controls, and the measures used to confirm the change held.

The AOD converter is the point at which carbon is removed from the melt and the final chemistry of the heat is set. Material additions, gas ratios and temperature windows differ by grade family, and the process runs continuously across three shifts. The recurring problem across all three pieces of work is the same one: operating knowledge sat with experienced individuals rather than in a controlled reference, so execution varied by shift and by operator.

The work was performed in an operations role. The documentation approach, meaning structured requirements, business rules, traceability and acceptance criteria, is the approach used in business analysis, applied here to a manufacturing process rather than a software system.

Document Control

Field	Detail
Prepared by	S. Satyam Singh
Role at the time	Associate Manager, AOD Operations
Business area	Steel Melting Shop, AOD Operations
Coverage	Grade-wise addition pattern standardisation; AOD auto batching requirements; 5S material allocation
Methods applied	Process mapping (As-Is / To-Be), requirements definition, business rules, DMAIC, RACI, control planning, validation scenarios, traceability
Grade families	200 series, 300 series, 400 series
Version	2.0 (portfolio edition)

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Part	Subject	Focus
A	Grade-Wise AOD Addition Pattern	Process mapping and standardisation of material addition sequence, temperature control and gas switch-over across three grade families
B	AOD Auto Batching	Business and functional requirements, bunker and weighing-hopper logic, exception handling and validation scenarios for automated material batching
C	5S Material Allocation at AOD	Current-state analysis, gap analysis and requirements for material location control, receipt handover and space utilisation

Terms Used

Term	Meaning
AOD	Argon Oxygen Decarburisation, the converter process used to refine stainless steel
Main blow / DB1 / DB2	Primary oxygen blowing stage, followed by second and third blow stages
Zero batch	Initial flux charge added before pouring
Dololime / Lime / Flux	Slag-forming additions used to control chemistry and protect refractory
Ferroalloys	Chromium, nickel and manganese bearing additions used to achieve target chemistry
MHS	Material Handling System: conveyors, bunkers, feeders and chutes serving the converter
CH1 / CH2	Charging hoppers feeding material into the converter
Weighing hopper	Intermediate hopper that weighs material before transfer to the charging chute
Reduction	Post-blow stage where chromium is recovered from slag

PART A · Grade-Wise AOD Addition Pattern

Process mapping and standardisation of material addition sequence, process controls and gas switch-over across three grade families.

A1. Background

AOD processing requires controlled addition of slag formers and ferroalloys at defined points in the blowing cycle. Each grade family requires a different addition pattern, flux quantity, alloy sequence, temperature window, stirring duration, reduction condition and gas switch-over point.

Before this work, grade-specific execution was carried out largely from operator experience and verbal handover. The addition sequence was not held in a single structured reference, and there was no consistent basis on which a new Shift-In-Charge could be brought up to competence.

A2. Business Problem

Area	Problem
Grade-wise variation	Each grade family required a different addition pattern, with no single reference covering all three
Process dependency	Operators depended on experience and verbal instruction rather than documented method
Flux control	Flux quantity differed by grade series and required a standard control point
Gas switch-over	Switch-over timing varied by grade and was not consistently documented
Temperature control	Sampling and reduction temperatures needed defined operating ranges
Chemistry achievement	Incorrect addition timing affected achievement of target chemistry
Shift consistency	Each shift required the same reference to produce comparable results
Safety	Enclosure closure and equipment readiness required clear, visible instruction

A3. Objectives

Objective	Description
Standardise addition pattern	Define the correct material addition sequence for each grade series
Improve process consistency	Ensure all shifts follow the same process flow
Improve productivity	Reduce process variation and avoid unnecessary delay in the blowing cycle
Improve chemistry control	Support control of the target element ranges for each grade
Improve gas control	Define switch-over timing by grade
Improve safety compliance	Document the safety points that apply to shop-floor execution
Support training	Enable less experienced team members to work to a defined grade-wise standard

A4. Scope

Area	Included
Grade families	Three series-level addition patterns
Grade-specific controls	Selected grades within each family requiring additional control
Flux addition	Main blow and subsequent blow stage flux addition
Alloy addition	Chromium, nickel and manganese bearing additions
Temperature control	Sampling and reduction temperature ranges
Gas switch-over	Switch-over timing and consumption reference
Safety	Enclosure closure and operational precautions

A5. Stakeholders

Stakeholder	Responsibility
Shift-In-Charge	Controls grade-wise execution during AOD operation
AOD Operator	Follows the defined addition sequence and process instructions
Production Manager	Monitors productivity, heat performance and shift compliance
Quality Team	Verifies chemistry achievement and grade-specific requirements
Safety Team	Ensures safe operation and shop-floor compliance
Maintenance Team	Ensures ladle, launder, chute and porous plug readiness
Gas supply coordination	Supports backup cooling requirement for specific grades

A6. As-Is and To-Be Process View

Area	As-Is	To-Be
Grade-wise process	Grade-specific instructions followed from experience	Standard process flow defined for each series
Addition sequence	Not held in one structured reference	Material addition sequence defined for each grade
Gas switch-over	Timing varied by grade, documented inconsistently	Switch-over condition defined per grade against a controlled range
Flux quantity	Different series required different quantities, applied from memory	Flux limits defined per series and process stage
Safety instruction	Held separately from operating practice	Safety points included in the operating reference
Training	New team members depended on senior guidance	Same reference available to all shifts and to new starters

A7. Process Maps

The maps cover the grade-wise flow from zero batch and pouring through main blow, flux addition, sampling, stirring and reduction. Each map covers one grade family and shows the operational sequence together with its process control window.

Map ID	Grade family	Trigger	Output
PM-01	300 series	Heat ready for AOD processing	Heat ready for reduction stage
PM-02	400 series	Heat ready for AOD processing	Heat ready for reduction stage
PM-03	200 series	Heat ready for AOD processing	Heat ready for reduction stage

A8. Standard Addition Pattern (Representative Sequence)

Sequence structure is as documented. Quantities and temperatures below are illustrative placeholders.

Step	Activity
1	Add zero batch at the series-defined flux quantity
2	Start pouring
3	Start process
4	Add slag former at the standard quantity for the series
5	Add chromium bearing alloy
6	Add flux at the series-defined interim quantity
7	Add nickel bearing alloy where the grade family requires it
8	Add metallic addition
9	Add manganese bearing alloy
10	Add flux up to the series maximum for main blow
11	Continue main blow
12	Add remaining flux during the subsequent blow stages
13	Take sample and temperature within the defined sampling window
14	Complete stirring phase for the defined minimum duration
15	Complete reduction within the defined reduction window and minimum duration

Series-level differences: the 200 series carries a higher interim and maximum flux quantity than the 300 and 400 series; the 300 series includes nickel bearing additions that the 400 series does not; the 400 and 200 series carry a defined minimum reduction duration.

A9. Grade-Specific Process Controls

Individual grades required additional controls beyond the family-level pattern. These were documented to reduce process variation on the grades most sensitive to execution.

Grade type	Key control points
Low-carbon austenitic	Chromium alloy added slowly ahead of the defined oxygen requirement. Flux not to be added while alloy addition is ongoing. Defined sampling and tapping temperature ceilings.
High-nickel austenitic	Backup cooling started before the heat. Tight upper limit on dissolved gas. Extra flux restricted. Ladle life, launder condition, slide gate plate, purging and slag control monitored.
Ferritic	Backup cooling arranged before the heat. Feed material availability confirmed. Dome and mouth cleaning completed before the heat. Both porous plugs confirmed working.
Low-nickel austenitic	Low build-up and controlled tapping sulphur. Opening phosphorus within range. Report checking and additive availability confirmed. Excess flux restricted.

A10. Gas Switch-over Control

Switch-over timing was defined grade by grade to control gas pick-up and improve process stability. At family level the pattern is as follows.

Grade family	Switch-over control
300 series	Timing varies by grade: ahead of the vacuum stage, during reduction, or in the second blow
400 series	Timing defined during main blow or reduction, depending on grade requirement
200 series	Timing linked to percentage completion of the reduction stage

A grade-level reference table was maintained pairing each grade with its controlled gas range and its switch-over trigger, expressed either as a percentage of main blow, a percentage of reduction, or a fixed offset from a process event. Actual ranges and triggers are omitted from this edition.

A11. Business Rules

Rule ID	Business rule
BR-A01	Each stainless steel series must follow its own defined AOD addition pattern
BR-A02	Zero batch must be added before pouring
BR-A03	Slag former must be added after process start, at the standard quantity
BR-A04	Flux addition must follow grade-wise quantity limits
BR-A05	Each series has a defined maximum flux quantity for the main blow stage
BR-A06	Remaining flux must be added during the subsequent blow stages
BR-A07	Sampling temperature must be maintained within the defined window
BR-A08	Reduction temperature must be maintained within the defined window
BR-A09	Stirring phase must meet the defined minimum duration
BR-A10	Where applicable to the series, reduction must meet the defined minimum duration
BR-A11	Gas switch-over must follow the grade-specific requirement
BR-A12	The converter enclosure must remain closed until tapping is complete
BR-A13	Excess flux must not be added for grades where it is restricted
BR-A14	Flux must not be added while alloy addition is ongoing
BR-A15	Charging hopper selection must be fixed before starting any grade sequence
BR-A16	Where opening silicon is high, flux quantity must be increased to process requirement

A12. Safety Requirements

Area	Control
Converter enclosure	Must remain closed until tapping is complete
Ladle condition	Ladle and launder must be clean before critical grades
Porous plug	Both porous plugs must be working where required
Chute condition	Reduction mixture chute must be clean
Purging	Purging must be sufficient to show bubbles at the surface
Mouth and dome	Dome and mouth cleaning completed before the heat where required
Personal protective equipment	Helmet, safety shoes, nose mask, hand gloves and jacket required for shop-floor execution

A13. Measures

Measure	Before standardisation	After standardisation
AOD process time	Variable between shifts	Controlled and consistent
Addition sequence	Experience-based	Reference-driven
Flux and alloy control	Risk of overlapping addition	Clearly separated by rule
Shift compliance	Dependent on individual practice	Measured against documented standard
Operator guidance	Verbal and experience-based	Documented step-by-step instruction
Process variation	Higher chance of variation	Reduced variation

PART B · AOD Auto Batching

Business and functional requirements, operating rules, exception handling and validation scenarios for automated material batching during the main blow window.

B1. Background

AOD operations require metallic and flux additions during the main blow period. Automated batching depends on accurate material selection, correct bunker configuration, appropriate feed and discharge rates, and clear trigger points tied to the oxygen phase.

For the sequence to run without interruption, material shifting, charging hopper movement, feeder condition, belt status, flame protection gates and auto-mode readiness must all be controlled before and during the process.

B2. Business Problem

The operational challenge was to control auto batching steps repeatably so that material additions complete within the expected main blow window. Without a structured process view, teams faced ambiguity around bunker priority, weighing hopper usage, colour status interpretation, feeder trips, charging hopper readiness and step table reconciliation.

Problem area	Business impact
Material addition timing	Delay in completing main blow additions and risk of process variation
Bunker and weighing hopper logic	Incorrect priority or simultaneous charging created operational confusion
Status visibility	Operators required clear colour-coded status to understand the material movement stage
Charging hopper control	Material must be fully discharged and tare weight zero before the next shift of material
Safety and exception handling	Feeder trip, converter tilting and power failure required clear control actions
Evidence and reconciliation	Step table output required tallying against the discharge table after addition

B3. Objectives

Objective
Document a standard process for AOD auto batching execution
Define business rules for bunker priority, material shifting, weighing hopper use and charging hopper readiness
Improve visibility by documenting colour-coded process status rules
Support completion of metallic addition and major flux addition within the main blow period
Reduce operational ambiguity between AOD, MHS, Electrical, Mechanical and shift teams
Create validation scenarios usable for process checks, training and system acceptance

B4. Scope

In scope	Out of scope
Auto batching process flow for all three grade families	Steel grade chemistry design and metallurgical formula ownership
Feed rate, discharge rate and material quantity requirements	Procurement and supplier commercial decisions
Bunker, weighing hopper and charging hopper operating rules	PLC coding and automation software development
Colour-coded status interpretation	Hardware design of conveyors, gates and feeders
Exceptions including feeder trip, converter tilting and power failure	Final approval authority for plant operating standards
Validation scenarios for process acceptance	Financial business case beyond process improvement logic

B5. Stakeholders

Stakeholder	Role in process	Requirement focus
Shift-In-Charge	Owns shift-level execution and decision making	Clear steps, exceptions, tally checks and safety controls
AOD Operator	Runs material addition steps and monitors the converter	Oxygen trigger points, charging hopper readiness, discharge status
MHS Team	Responsible for material handling equipment and feeders	Feeder health, belt status, bunker movement, chute availability
Electrical Team	Supports electrical readiness and flame protection gates	Auto mode, gate health, power-related exception response
Mechanical Team	Supports converter positioning and mechanical issues	Cooling position, back pressure checks, emergency handling
Production Management	Tracks productivity and process performance	Main blow timing, process discipline, shift compliance
Quality / Process Team	Checks grade requirements and output stability	Controlled chemistry and consistency of material addition

B6. Step Table Structure

Structure as documented. All values below are illustrative placeholders and do not represent operating parameters.

Step	Bunker	Trigger	Material	Qty	Feed rate	Discharge rate	Shift time	Remarks
S0	B-a	Prior heat, third blow	Slag former	Q1	n/a	n/a	0	Shift during previous heat
S1	B-b	Early oxygen band	Slag former	Q1	n/a	n/a	0	Shift when pouring starts; hold reserve on charging chute
S2	B-c & B-d	Oxygen band 2	Chromium alloy	Q2	F1	D1	T1	Shift as soon as confirmed
S3	B-e & B-f	Oxygen band 3	Metallic addition	Q3	F2	D2	T2	
S4	B-g & B-h	Oxygen band 4	Manganese alloy	Q4	F1	D1	T3	
S5	B-i	Oxygen band 5	Flux	Q5	F3	D3	T4	
Total	Expected total process time reference, compared against the main blow window							

A separate step table was maintained for each of the three grade families. The 300 series table carries additional nickel-bearing steps and therefore a longer expected total; the 400 series carries the shortest.

B7. As-Is and To-Be Process View

Area	As-Is observation	To-Be requirement
Process reference	Operational rules held across notes, step tables and team knowledge	A single structured process document for execution and training
Material movement	Material shifting depended on correct understanding of bunker and weighing hopper rules	Defined priority rules and conditions for simultaneous or sequential bunker operation
Status understanding	Operators interpreted colour-coded states from experience	Each status colour and its meaning documented
Safety controls	Critical exceptions handled through SOP and shift experience	Exception rules included in the process control section
Validation	Step table and discharge table comparison required after addition	Reconciliation defined as a formal post-addition control

B8. Functional Requirements

ID	Area	Requirement	Priority
FR-01	Series selection	The process shall support auto batching step tables for each grade family	High
FR-02	Material setup	The process shall define material name, quantity, bunker, feed rate and discharge rate for each step	High
FR-03	Oxygen trigger	Each material addition step shall be linked to a defined oxygen trigger or process event	High
FR-04	Colour status	The process shall follow colour-coded status for filling, shifting, ready or discharging, and completed states	High
FR-05	Bunker priority	Where multiple bunkers use the same weighing hopper in one step, a fixed left-to-right priority shall be followed	High
FR-06	Parallel bunker use	Where bunkers use different weighing hoppers in one step, simultaneous start shall be allowed	Medium
FR-07	Charging hopper control	Material shall shift only after the previous material is fully discharged and tare weight is zero	High
FR-08	Belt status	Material shall shift only when belt status is green; red status shall block shifting	High
FR-09	Auto mode	Equipment required for auto batching shall remain in auto mode unless exception handling is active	High
FR-10	Post-addition check	After addition, the step table shall be tallied against the discharge table	High
FR-11	Tilting control	During converter tilting in main blow, both charging hoppers shall be discharge-stopped before tilting	High
FR-12	Emergency handling	During power failure or critical interruption, additions shall stop and emergency checks shall be initiated	High

B9. Business Rules

Rule ID	Business rule
BR-B01	Metallic addition and major flux addition must be completed within the main blow planning window
BR-B02	A dedicated bunker is allocated to the primary slag former as part of the auto batching configuration
BR-B03	Bunkers not required for a specific grade family must be marked out of service
BR-B04	Status colour 1 means weighing hopper filling is in progress or not yet complete
BR-B05	Status colour 2 means material is shifting to the charging chute
BR-B06	Status colour 3 means all material has shifted to a charging hopper
BR-B07	Status colour 4 means material is ready to discharge or is discharging
BR-B08	Status colour 5 means the step is complete
BR-B09	For the same weighing hopper, bunker priority follows a fixed sequence and must not start simultaneously
BR-B10	For different weighing hoppers, bunkers may start simultaneously within the same step
BR-B11	The feeder must not trip during charging or discharging; MHS owns feeder readiness
BR-B12	The outlet flame protection gate must be healthy and inlet gates manually open where required
BR-B13	After zero batch is shifted to a charging hopper, the empty source must be marked out of service
BR-B14	On main blow power failure, the lance emergency control must be operated so the lance returns to rest position

B10. Data Requirements

Data field	Purpose
Series	Identifies the applicable grade family step table
Step	Defines sequence number for the batching step
Bunker	Identifies source bunker for material
Oxygen trigger	Defines process event or oxygen trigger for addition
Material name	Defines material to be added
Material quantity	Defines target material quantity
Feed rate	Defines material feeding rate
Discharge rate	Defines belt discharge rate
Time to shift chute	Calculates expected shift time
Remark	Captures execution condition or operational note

B11. Non-Functional Requirements

ID	Category	Requirement
NFR-01	Safety	The process must prevent unsafe material shifting during converter tilting, power failure or equipment abnormality
NFR-02	Usability	Status colours and step tables must be readily interpretable by operators and Shift-In-Charges
NFR-03	Traceability	Step table and discharge table must be reconcilable after addition
NFR-04	Reliability	Feeder, gates and auto-mode readiness must be verified to reduce process interruption
NFR-05	Maintainability	Business rules must be straightforward to update when grade, bunker or material configuration changes

B12. Exception Handling and Control Plan

Exception	Immediate control	Owner	Evidence
Feeder trip during charging or discharging	Stop, check feeder health and resolve before continuing material movement	MHS Team	Equipment status and shift log
Belt shows red status	Do not shift material until status is green	AOD Operator / MHS	Status indication
Material not fully discharged	Do not shift next material until discharge is complete and tare weight is zero	AOD Operator	Tare weight check
Converter tilting during main blow	Both charging hoppers must be discharge-stopped before tilting	Shift-In-Charge / AOD Operator	Operator confirmation
Power failure during blowing	Stop all additions, confirm emergency gas valve, call Electrical and Mechanical, move converter to cooling position as required	Shift-In-Charge / Electrical / Mechanical	Power failure response log
Main blow power failure	Operate the lance emergency control so the lance returns to rest position	AOD Operator	Emergency action confirmation

B13. Validation Scenarios

ID	Precondition	Action	Expected result
V-01	A grade family is selected	Run that family's step table	Sequence follows S0 to the final step and total time falls within the expected reference
V-02	300 series selected	Run the 300 series step table	Nickel, chromium, metallic, manganese and flux steps follow the defined sequence
V-03	Same weighing hopper used by multiple bunkers	Trigger all three bunkers in the same step	The first starts, then the others in sequence; simultaneous start is blocked
V-04	Different weighing hoppers used	Trigger all three bunkers in the same step	All permitted bunkers may start simultaneously
V-05	Belt status red	Attempt material shift	Material shift is not allowed
V-06	A charging hopper holds a material balance	Attempt next material shift	Process waits until material is fully discharged and tare weight is zero
V-07	Converter tilting during main blow	Initiate tilting control	Both charging hoppers are discharge-stopped before tilting
V-08	Power failure during blowing	Apply emergency process	Additions stop, emergency gas valve is checked, Electrical and Mechanical are called
V-09	Addition complete	Compare step table with discharge table	Material quantities and steps reconcile

B14. Requirements Traceability

Requirement	Linked rule or control	Validation	Business value
FR-01	Series-wise step tables	V-01, V-02	Supports grade-wise process control
FR-02	Material setup table	V-01, V-02	Ensures quantity and material detail are controlled
FR-04	Colour status rules	V-05	Improves operator visibility
FR-05	Same weighing hopper priority rule	V-03	Controls sequential bunker operation
FR-06	Different weighing hopper simultaneous rule	V-04	Controls parallel bunker operation
FR-07	Tare weight and discharge completion	V-06	Prevents incorrect material movement
FR-10	Step table against discharge table tally	V-09	Creates post-addition evidence
FR-11	Tilting control	V-07	Controls safety risk
FR-12	Emergency handling	V-08	Controls power failure response

B15. RACI Matrix

Activity	Shift-In-Charge	AOD Operator	MHS	Electrical	Mechanical	Production
Confirm grade and series step table	A	R	C	I	I	C
Verify material and bunker readiness	A	R	R	I	I	C
Check auto mode and flame gates	A	R	C	R	C	I
Run material shifting and discharge steps	A	R	R	I	I	I
Handle feeder trip	A	C	R	C	C	I
Handle power failure	A	R	C	R	R	I
Tally step table and discharge table	A	R	C	I	I	C

R = Responsible · A = Accountable · C = Consulted · I = Informed

PART C · 5S Material Allocation at AOD

Current-state analysis, gap analysis and business requirements for material location control, receipt handover and space utilisation in the AOD material allocation area.

C1. Background

This work addressed 5S implementation for material allocation in the AOD area of the Steel Melting Shop. The starting requirement was to improve visual management and reduce material wastage. Delivery covered process review, stakeholder discussion, material location tagging, bin allocation, receipt handover control and improvement of material handling practice.

The analysis required understanding the current operating process, identifying pain points, capturing stakeholder requirements, defining business rules and documenting the future process for shop-floor execution.

C2. Business Problem

Problem area	Business problem
Material receipt delay	Receipt information could carry over between shifts, creating confusion at handover
Visual management	Material locations were not readily identifiable from the control room or shop floor
Space utilisation	Long-stored material and unplanned placement reduced available working space
Small quantity material handling	Low-volume additives required better sub-bin allocation
Communication gap	Melters, operators, technicians and helpers needed a clear location reference for each material
Grade-wise equipment readiness	Chute requirement varied by grade family, so availability needed to match the running heat plan
Ownership gap	Material unloading and placement required clear responsibility, particularly on night shift

C3. Objectives

Objective	Description
Improve visual management	Create clear identification and tagging for material locations
Reduce material wastage	Improve material allocation discipline and reduce incorrect placement
Improve shift handover	Ensure receipt and material information is available before the next shift takes over
Improve material handling	Define bins, sub-bins and standard locations for frequently used materials
Improve communication	Allow the control room to communicate material location clearly to technicians and helpers
Improve process control	Align material booking and chute availability with the heat plan and grade requirement
Sustain the improvement	Create controls so the improved process can be maintained over time

C4. Scope

In scope	Out of scope
Material allocation in the AOD area	Supplier commercial negotiations
Tagging and nomenclature for material locations	Change to metallurgical grade design
Bin and sub-bin division for small quantity materials	Major civil construction work
Receipt handover and material identification control	ERP or software implementation
Material booking alignment with the heat plan	Automation of the weighing bridge system
Chute availability by grade requirement	Plant-wide 5S implementation outside AOD

C5. Stakeholders

Stakeholder	Need or interest	Engagement approach
Production Manager	Improved productivity and reduced wastage	Establish business priority and success measures
Shift-In-Charge	Clear material availability and handover information	Capture shift-level pain points and execution needs
Melter / Operator	Quick identification of required material	Understand real user workflow from the control room
Technician / Helper	Clear physical location of materials	Capture location communication issues
Store / Bunker Yard Team	Correct unloading and allocation reference	Define receipt and placement coordination
Material Unloading Supervisor	Ownership of unloading and correct placement	Define responsibility and handover control
Maintenance / Mechanical Team	Space allocation and equipment placement	Capture constraints on shop-floor movement and space

C6. Method

The work followed a DMAIC structure, with the analysis and documentation produced at each phase set out below.

Phase	Activity	Output
Define	Established the need to perform 5S of material allocation at AOD to improve visual management and reduce wastage	Problem definition, scope and objectives
Measure	Identified the need for nomenclature, bin division and a material allocation mechanism	Current state review, data points, process observation
Analyse	Reviewed current trends and shop-floor issues affecting material allocation	Gap analysis, root cause review, stakeholder discussion
Improve	Proposed material handling improvement, tagging, sub-bins and receipt control	To-Be process design, business requirements, options
Control	Defined how 5S at AOD would be implemented and maintained	Control plan, ownership, KPIs, process governance

C7. Pain Points and Gap Analysis

Pain point	Operational impact	Required change
Receipt handover delay	Shift-to-shift information delay created confusion over material receipt	Increase receipt copies and include shift, timing and chemistry detail
Unclear location identification	Operators and helpers could not identify material locations quickly	Tag all material locations with permanent, visible labels
No sub-bin system for small materials	Low-volume additives were harder to locate and control	Divide bins into sub-bins for low-volume additives
Material held for long periods	Space remained blocked by slow-moving material	Clear long-stored material gradually and set maximum quantity limits
Imprecise material booking	Excess or incorrect material availability affected space and planning	Book material from stores and bunker yard according to the heat plan
Chute availability not grade-aligned	Unnecessary chutes occupied space while required chutes could be unavailable	Vary chute availability based on the running grade family
Incorrect material placement	Material placed at the wrong location caused execution issues	Assign a material unloading supervisor with material knowledge

C8. Business Requirements

ID	Business requirement	Priority
BR-C01	The AOD material allocation area must have clearly tagged locations for each identified material category	High
BR-C02	Material bins must be divided into sub-bins for low-volume additives	High
BR-C03	Filled material trucks must carry an additional receipt copy to support shift handover	High
BR-C04	Receipt detail must include shift, timing and chemistry detail where applicable	High
BR-C05	Material must be booked from stores and bunker yard precisely according to the heat plan	High
BR-C06	Chute availability must be adjusted to the running grade family	Medium
BR-C07	Slow-moving materials must be cleared gradually to release space	Medium
BR-C08	Quantity of bulk material held in the area must be controlled to avoid unnecessary space blockage	Medium
BR-C09	Standard weights used for weighing bridge calibration must be tagged	Medium
BR-C10	A material unloading supervisor must be available in the receiving bay to ensure correct placement	High

C9. Functional Requirements

ID	Functional requirement
FR-C01	The material allocation process must support location identification from the control room using tagged location references
FR-C02	The process must allow operators to communicate material location to technicians and helpers without ambiguity
FR-C03	The bin structure must support separate storage of small quantity materials
FR-C04	The receipt handover process must ensure material receipt status is available before shift handover
FR-C05	The material unloading process must include verification by a responsible person before final placement
FR-C06	The process must support grade-wise chute planning based on the running heat plan

C10. Business Rules

Rule ID	Business rule
BU-C01	Each material category must be stored only in its identified tagged location
BU-C02	Small quantity materials must be placed in defined sub-bins and not mixed with bulk materials
BU-C03	Material receipt handover must include receipt count, shift, unloading time and chemistry information where applicable
BU-C04	Material must be booked according to the heat plan rather than to excess availability
BU-C05	Chute availability must match the grade plan for the running series
BU-C06	Chute requirement must be reduced where the running series permits, to avoid unnecessary space usage
BU-C07	Long-stored materials must be reviewed and cleared gradually to create space
BU-C08	The material unloading supervisor must ensure material is not placed at the wrong location
BU-C09	Tagging must be visible enough for store, bunker yard and AOD team members to identify locations

C11. Acceptance Criteria

ID	Acceptance criterion
AC-01	Given a material truck is unloaded, when shift handover happens, then the next shift receives clear receipt information without confusion
AC-02	Given a melter or operator requests a material location, when the control room checks the tagged layout, then the location can be communicated clearly to the technician or helper
AC-03	Given small quantity materials are received, when they are stored, then they are placed in the correct sub-bin
AC-04	Given a heat plan is available, when material is booked from stores or bunker yard, then booking matches the heat plan requirement
AC-05	Given a grade family is running, when chutes are arranged, then only the required number of chutes occupies the area
AC-06	Given a 5S audit is performed, when the material area is reviewed, then material tags, bin allocation and space clearance are visible and traceable

C12. RACI Matrix

Activity	Analyst	Shift-In-Charge	Store / Bunker Yard	Unloading Supervisor	Production Manager
Capture current process and pain points	R	C	C	C	A
Define material tagging requirement	R	C	C	C	A
Prepare bin and sub-bin allocation logic	R	C	C	C	A
Implement receipt handover control	C	A	R	R	C
Ensure correct material placement	C	A	C	R	C
Monitor 5S sustainment	C	R	C	R	A

R = Responsible · A = Accountable · C = Consulted · I = Informed

C13. Measures

Measure	Measurement method	Target
Material location identification time	Time taken to identify material location from control room or shop floor	Reduced search time
Incorrect placement incidents	Count of materials found in an incorrect location	Reduced incidents
Receipt handover accuracy	Receipt count and information available during shift handover	Improved handover accuracy
Tagged location completion	Number of material locations with visible tags	All critical locations tagged
Sub-bin readiness	Availability of sub-bins for small quantity materials	Defined sub-bins available
Space availability	Review of space blocked by slow-moving material	Improved space usage
5S compliance score	Periodic 5S audit of the AOD material allocation area	Sustained compliance

C14. Benefits

Area	Benefit
Visual management	Material locations become easier to identify for the AOD team, store and bunker yard
Communication	Melter or operator can direct technician and helper using clear location references
Shift handover	Receipt information becomes clearer between shifts
Space management	Long-stored and excess material can be controlled to improve working space
Material control	Small quantity materials are separated and easier to track
Operational efficiency	Less time lost searching for, clarifying and correcting material placement
Sustainment	Control measures support maintenance of 5S after implementation

Confidentiality. The employer is not named in this edition. All quantities, temperatures, gas ranges, equipment identifiers, feed and discharge rates, timings and grade codes have been removed or replaced with illustrative placeholders. What is shown is the analysis structure, requirement and rule logic, control design and validation approach.